Bio-Image analysis Basics - ImageJ/FIJI

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Bridging Nordic Microscopy Infrastructures
SciLifeLab
National Microscopy Infrastructure
EURO-BIOIMAGING

Bio image analysis

We will use ImageJ, and in particular FIJI to illustrate the most basic concepts of bio-image Analysis

ImageJ - https://imagej.net/ij/index.html

FIJI - https://imagej.net/software/fiji/

Check installation!

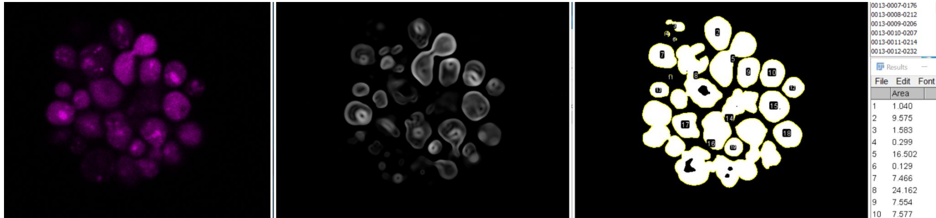
home | news | docs | download | plugins | resources | list | links

ImageJ

Image Processing and Analysis in Java

Search

Fiji Is Just ImageJ



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Program

- Images are numbers
- Math over pixels and images
- Measurements on images
- Pre-processing
  - Image filters
- Segmentation
  - Post processing - morphological operations
- Quantification
  - Analyze particles, ROI manager

Workflow

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Important resources

Basics of Bio-Image Processing and Analysis

ImageJ/FIJI: https://imagej.net/learn/index

NEUBIAS: https://neubias.github.io/training-resources/

Image.sc Forum https://forum.image.sc/

Bio-Image analysis wiki https://wiki.cmci.info/

Books:

Bioimage Data Analysis, Editor: Kota Miura
https://analyticalscience.wiley.com/do/10.1002/was.00050003

Bioimage Data Analysis Workflows, Editors: Kota Miura and Nataša Sladoje
https://www.springer.com/gp/book/9783030223854

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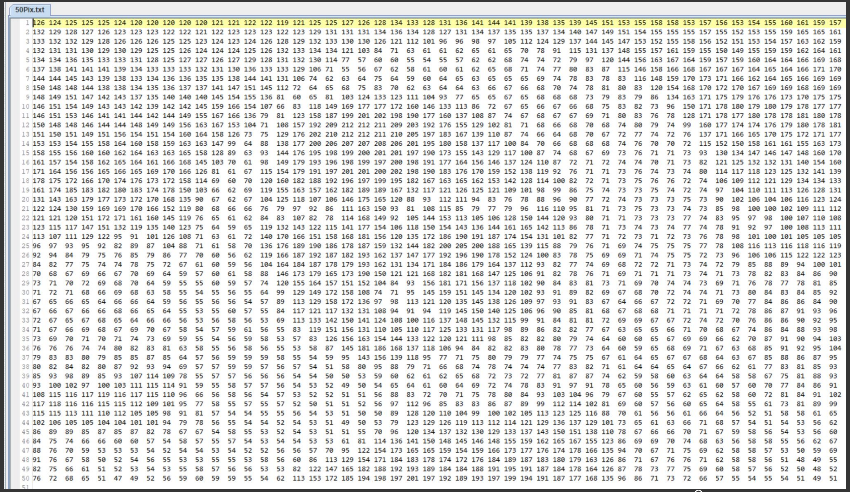
1 Images are numbers

1.1 Do you recognize this biological structure?

File → Open → 50pix.txt (or drag and drop)

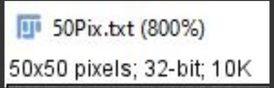
File → Import → Text Image → 50pix.txt



1.2 What is bit-depth?

Try: Analyze → Histogram,

Look at the top left corner of your image



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The Mona Lisa image used comes from wikimedia commons:
https://commons.wikimedia.org/wiki/File:Mona_Lisa,_by_Leonardo_da_Vinci,_from_C2RMF_retouched.jpg

1 Images are numbers

1.3 Image formats and meta-data

File → Open → Gray.tif (or drag and drop)

Look at the top left corner of your image window, what is the bit depth? Why?

1.4 RGB images, Channels and Composite image

File → Open → RBG.tif

Image → Duplicate → Title: Channels | Important!

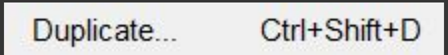


Image → Type → RGB stack | What happens to the image - multidimensional!

Image → Color → Make composite

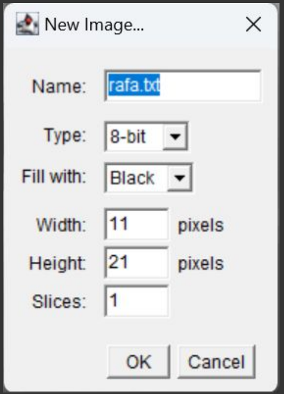
Detour 1 - export .gif

1 Images are numbers

1.5 Creating your own image

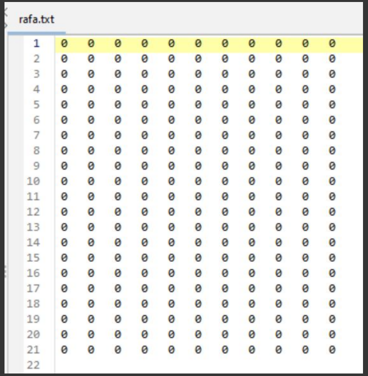
File → New → Image

Use for example:
your-name.txt, type: 8-bit, Fill with: black, Width: 11, Height: 21, Slices: 1



File → Save As→ Text Image (make it end in .txt)

File → Open (the .txt file)



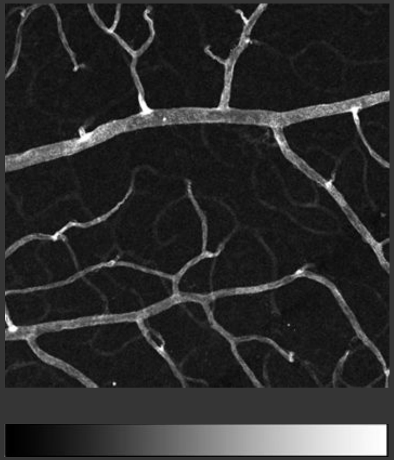
Change values in the text editor (0-255), save and import again.

Detour: let us discuss about bit depths

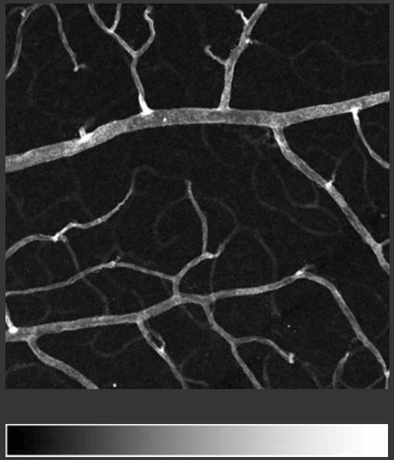
1 Images are numbers

Bit depth: **number of bits used to represent the intensity value of each pixel, determining the range of measurable signal levels (dynamic range).**

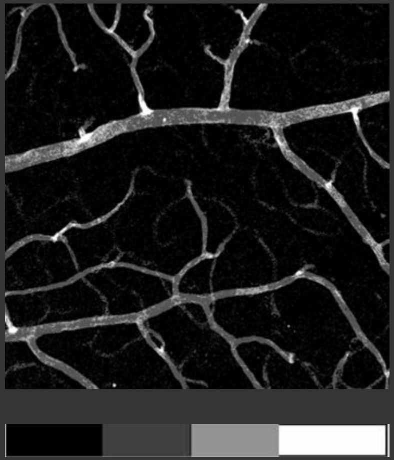
16 bits: 65536 steps



8 bits: 256 steps



2 bits: 4 steps



1 bits: 2 steps

1 Colors and channels

1.6 Composite image, image stacks!

File → Open Samples → Neuron

Image → Color → Make Composite → Composite: ok

What happens to the image

1.7 Channels tools

Image → Color → Channels tool

Play with the options, Try to get the histogram again

1 Conversions

1.8 Bit depth conversion

File → Open Samples → Neuron

Image → Duplicate → Channel 2, title: green

Image → Adjust → Brightness/Contrast

Try with Auto, Reset, and Apply [DANGER]

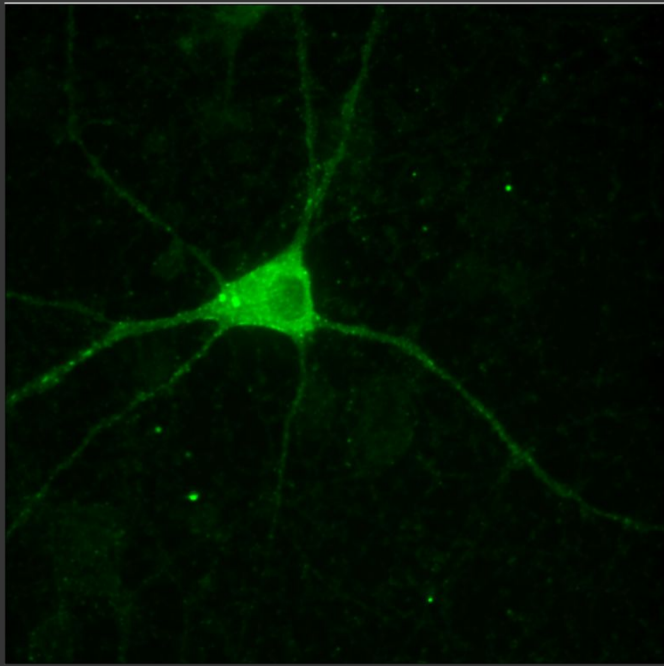
Image → Type → 8 bit

Look at the histograms

Edit → Options → Conversions, Try without scale activated

Discuss gamma correction, and CLAHE

Process → Math → Gamma



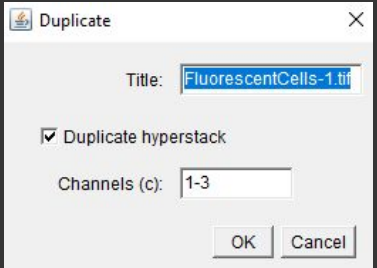
1 Colors and channels

1.9 Stacks to images and images to stack

File → Open samples → Fluorescent Cells

Image → Stacks → Make montage → ok

Image → Duplicate: Note the differences when using stacks



Draw a rectangle, Image → Duplicate

Draw a rectangle, Image → Crop

Image → Stacks → Stacks to images

Image → Colors → Merge channels

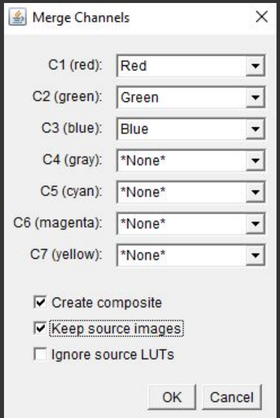


Image → Stacks → Tools → Concatenate: Are these channels?

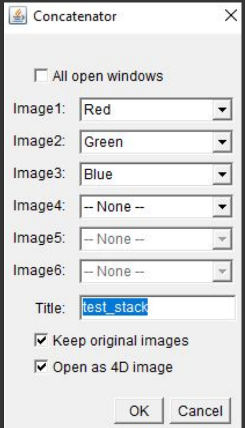
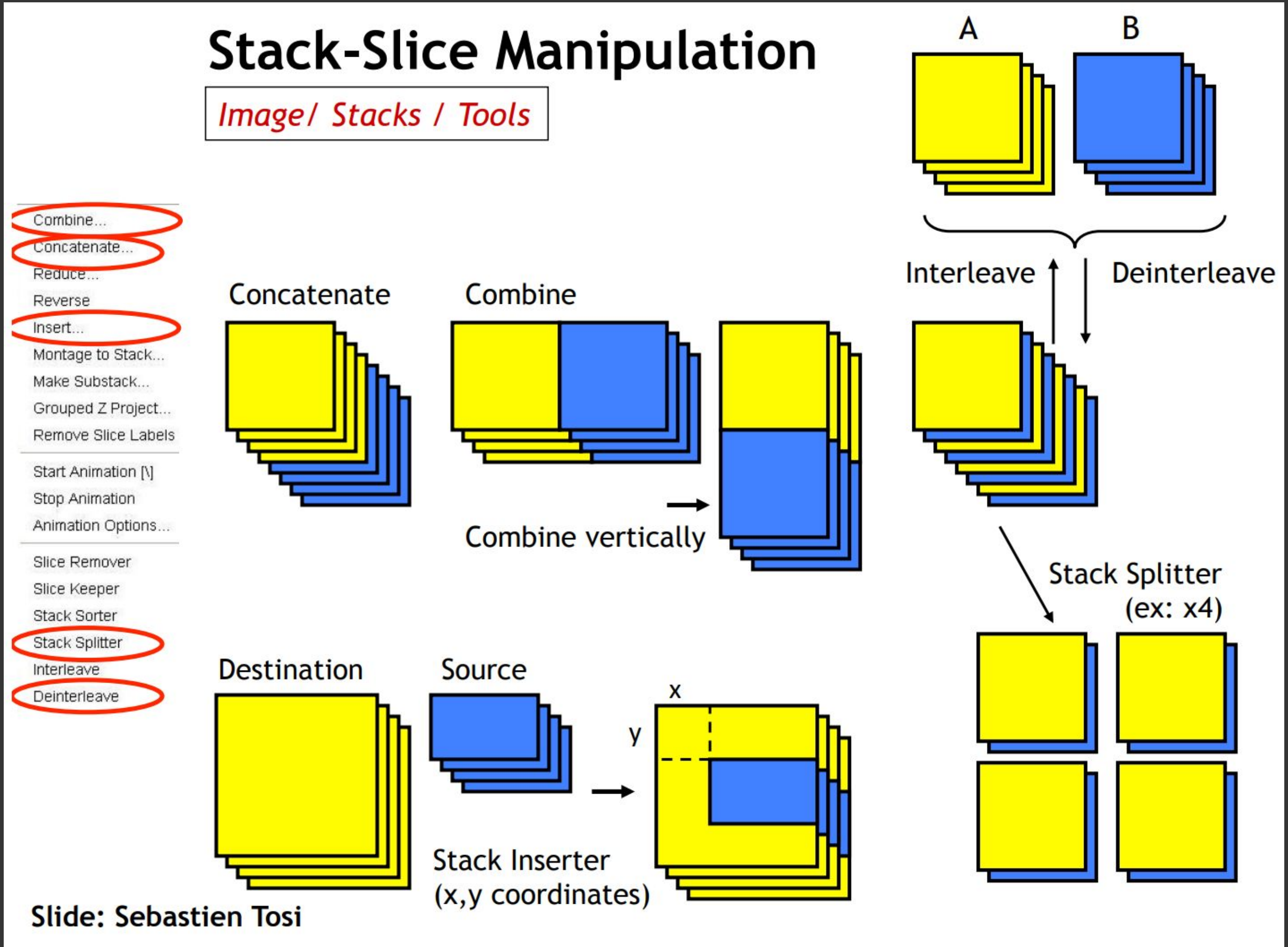
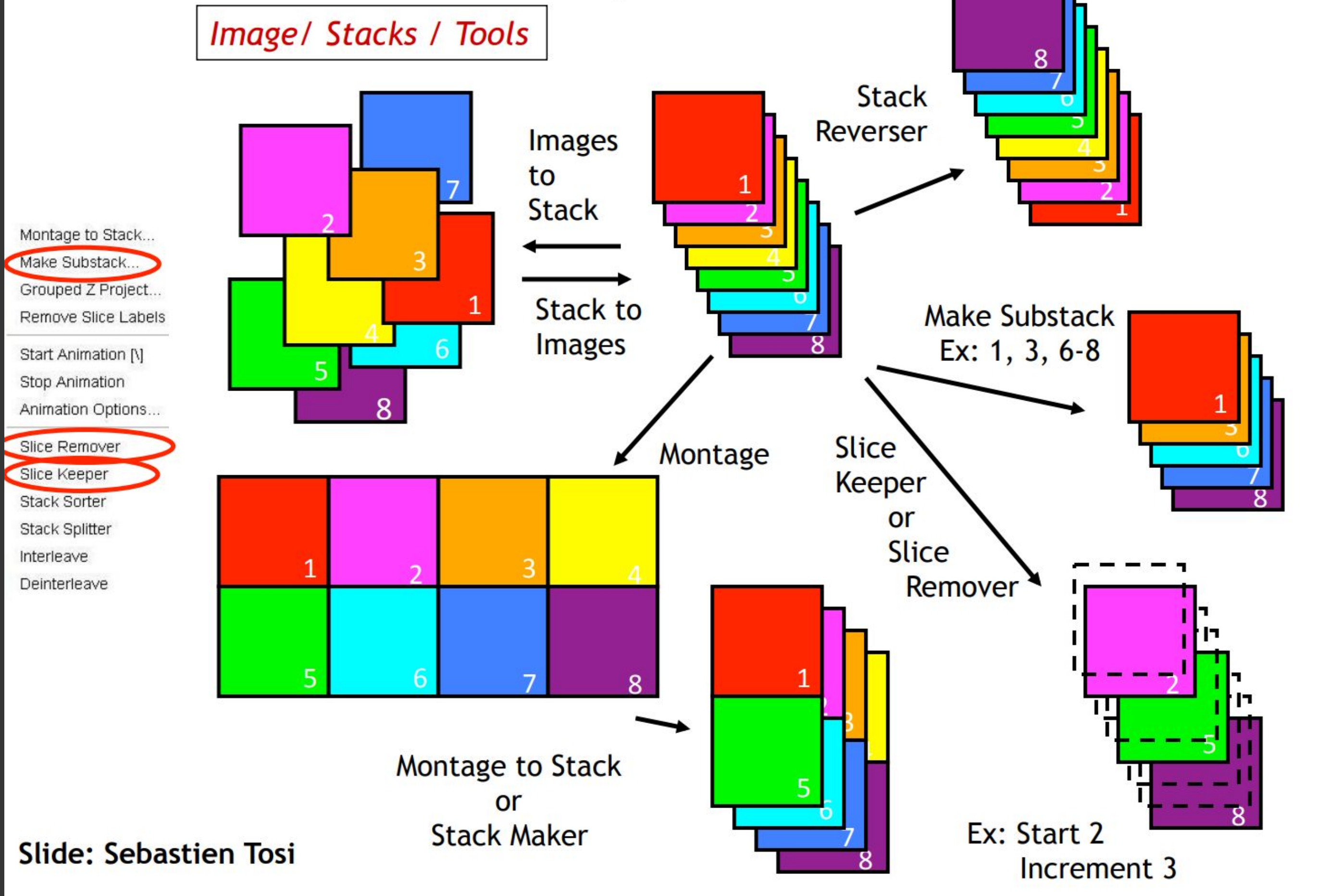


Image → Hyperstacks→ Stack to Hyperstack // Reorder Hyperstack


Stack-Slice Manipulation
Image/ Stacks / Tools
Combine...
Concatenate...
Reduce...
Reverse
Insert...
Montage to Stack...
Make Substack...
Grouped Z Project...
Remove Slice Labels
Start Animation [\]
Stop Animation
Animation Options...
Slice Remover
Slice Keeper
Stack Sorter
Stack Splitter
Interleave
Deinterleave
Concatenate
Combine
Combine vertically
A
B
Interleave
Deinterleave
Stack Splitter
(ex: x4)
Destination
Source
x
y
Stack Inserter
(x,y coordinates)
Slide: Sebastien Tosi

Stack-Slice Manipulation



Slide: Sebastien Tosi

File → Close all

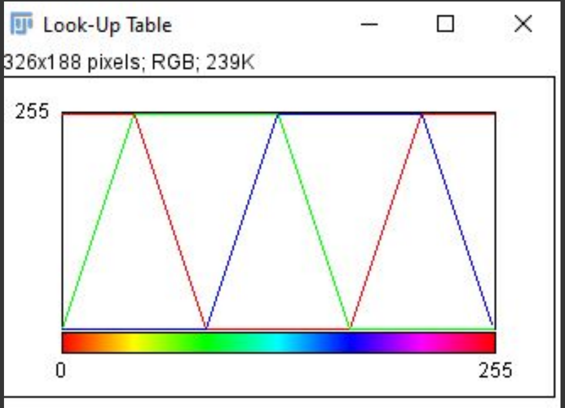
1 Colors and channels

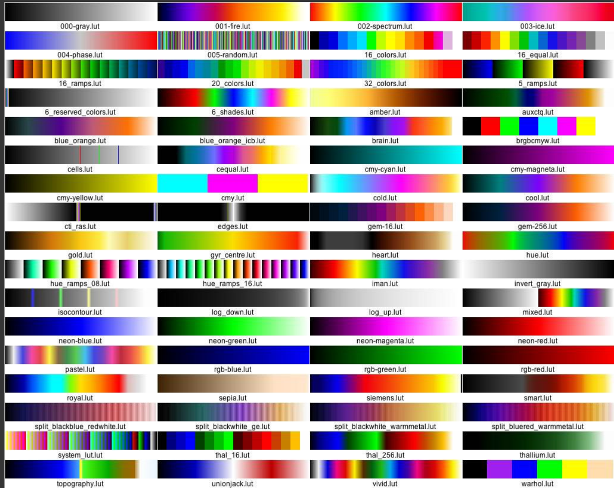
1.10 LUTs

File → Open samples → Fluorescent Cells

Navigate to blue channel, change LUT: Image → LookUp Tables

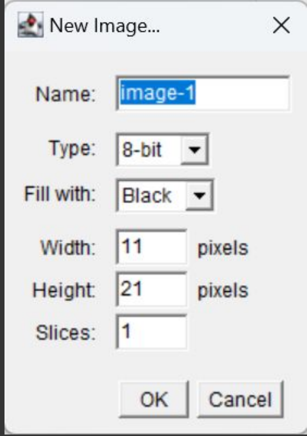
Image → Color → Show LUT



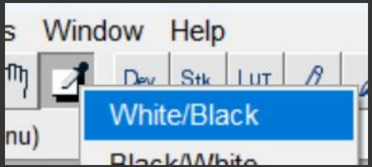


2 Math

2.1 Basic operations

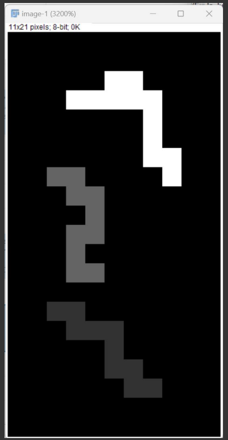


File → New → Image (image-1, 8bit, black, 11x21 pix)



Color picker → right-click → white/black

Pencil tool



Color picker → double-click → change value (<255) → Pencil tool

Process → Math → Add (try others, use preview and apply)

What happens to max/min values? How division and multiplication works?

Change to 32 bits, explore again +, -, /, *

2 Math

2.2 Operations between images

File → New → Image (image-1, 8bit, black, 11x21 pix)

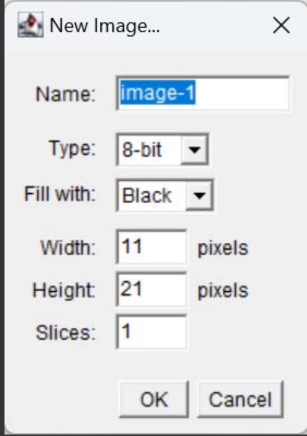
File → New → Image (image-2, 8bit, black, 11x21 pix)

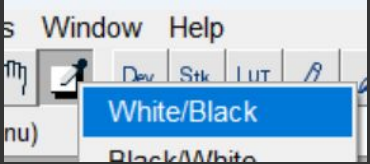
Draw 2 different patterns, use different values between 0-255

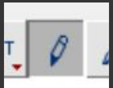
Process → Image Calculator

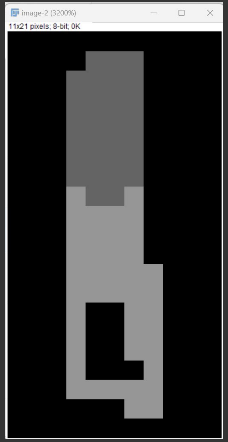
Try for example Addition, new image; then 32bit output

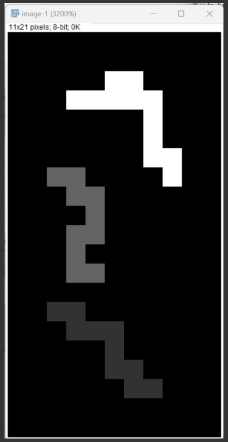
Try: What happens if the images are of different size?





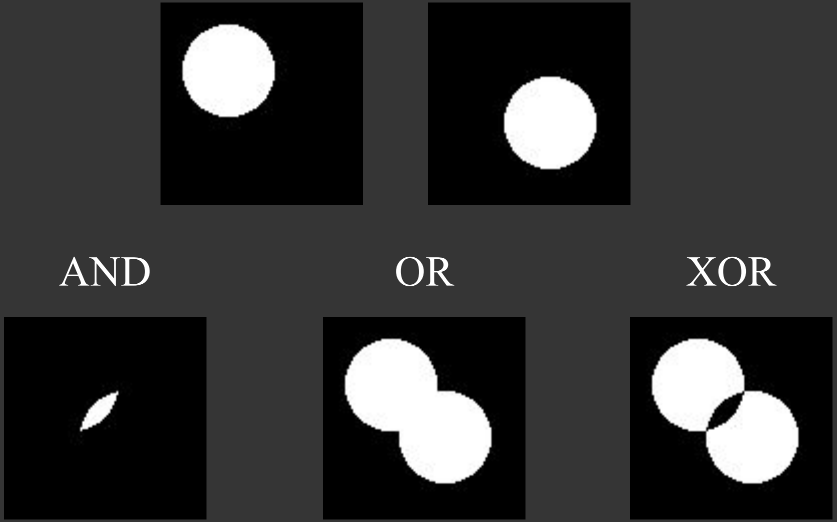






2 Math

2.2 Operations between images



Add	img = img1+img2
Subtract	img = img1-img2
Multiply	img = img1*img2
Divide	img = img1/img2
AND	img= img1 AND img2
OR	img = img1 OR img2
XOR	img = img1 XOR img2
Min	img = min(img1,img2)
Max	img = max(img1,img2)
Average	img = (img1+img2)/2
Difference	img = |img1-img2|

2 Math

2.3 Image thresholding - segmentation

If pixel_value > Th then new_val = true, else new_val = false

File → Open samples → Fluorescent Cells

Image → Duplicate → Channel 0, use name “red”

Image → Adjust → Threshold [Otsu, see Th value, then apply]

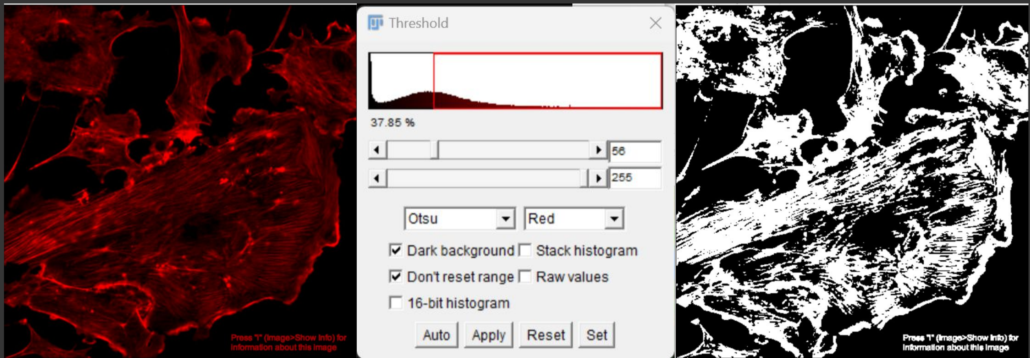
Edit → Selection → Create Selection

Analyze → Set Measurement → Area

Analyze → Measure

Edit → Selection → Select None

Analyze → Measure



2 Math

2.4 Image thresholding - overlap

File → Open samples → Fluorescent Cells

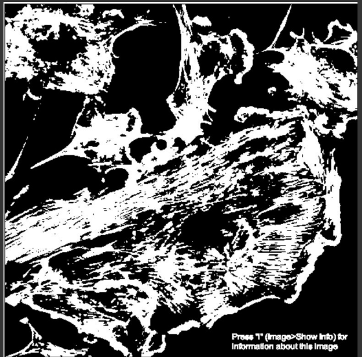
Image → Duplicate → Channel 0, use name “red”

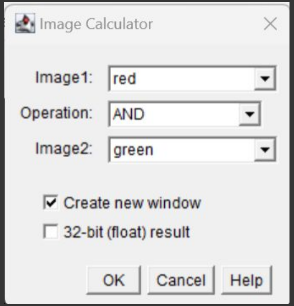
Image → Duplicate → Channel 1, use name “green”

Image → Adjust → Threshold [Otsu, see Th value, then apply] Do this on both images

Process → Image Calculator → Use AND

Can you measure overlap?





—

2 Math

2.5 Image scale

File → Open samples → Hela Cells & Fluorescent cells

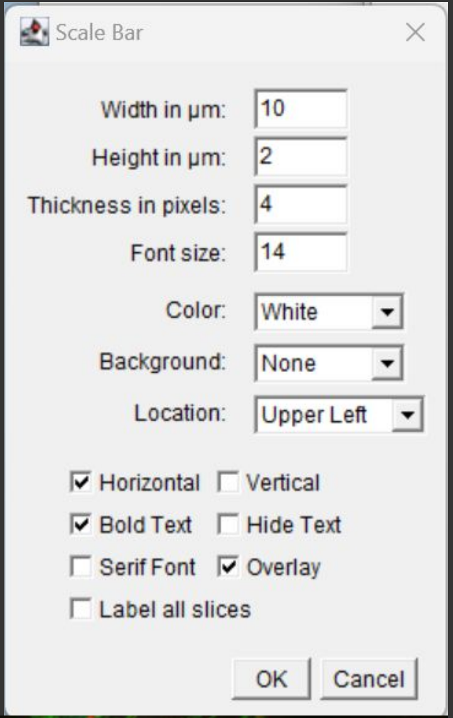
Press “i”, to see info

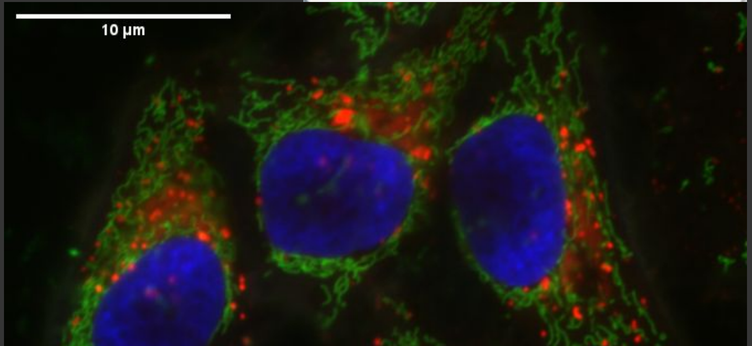
Image → Properties | Which of the images has information about scale?

Analyze → Tools → Scale Bar…

Image → Type → RGB Color

File → Save As → PNG… → <name>.png





3 Measurements

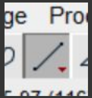
3.1 Measure lengths

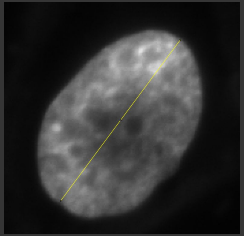
File → Open samples → Hela Cells

Image → Duplicate (Channel 3, call it nuc)

Image → Lookup Tables → Gray

Use line profile tool to draw



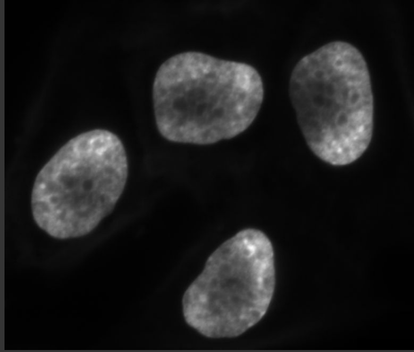


Analyze → Measure

Image → Properties → Change pixels to 1,1,1 no units (-)

Analyze → Measure

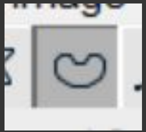
3 Measurements



3.2 Measure area

File → Open samples → Hela Cells
Image → Duplicate (Channel 3, call it nuc)
Image → Lookup Tables → Gray

Use freehand selection tool



Analyze → Set Measurements → Area
Analyze → Measure

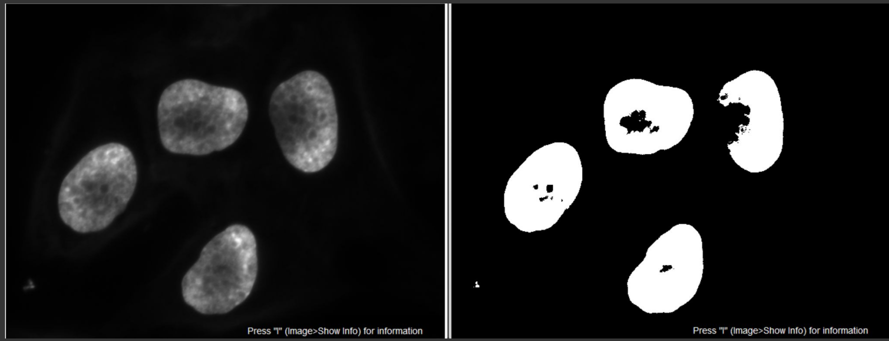
Image → Adjust → Threshold [Otsu, apply]
User wand tool, select and object

Analyze → Set Measurements → Area
Analyze → Measure

3 Measurements

3.3 Measure Intensity

File → Open samples → Hela Cells
Image → Duplicate (Channel 3, call it nuc)
Image → Lookup Tables → Gray
Image → Duplicate (Call it nuc-seg)
Image → Adjust → Threshold [Otsu, apply]
Analyze → Set Measurements → Mean gray value
Wand → Select segmented object
Click on nuc Image
Edit → Selection → Restore selection
Analyze → Measure



4 misc - Plugins and update sites

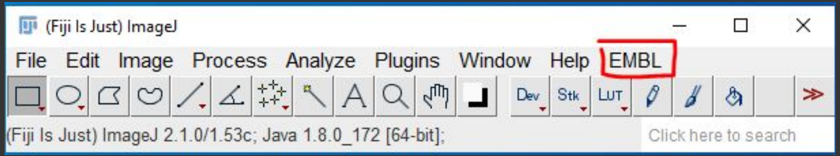
4.1 Installation of plugins listed in update sites

Go to: https://imagej.github.io/list-of-update-sites/

Lets us install the plug-in CMCI-EMBL

Help → Update → Manade update sites → CMCI-EMBL → Close → Apply Changes

After restart you should now see the EMBL tab:



3 Measurements

3.4 Measure Counts

EMBL→ Samples → Quantum dots

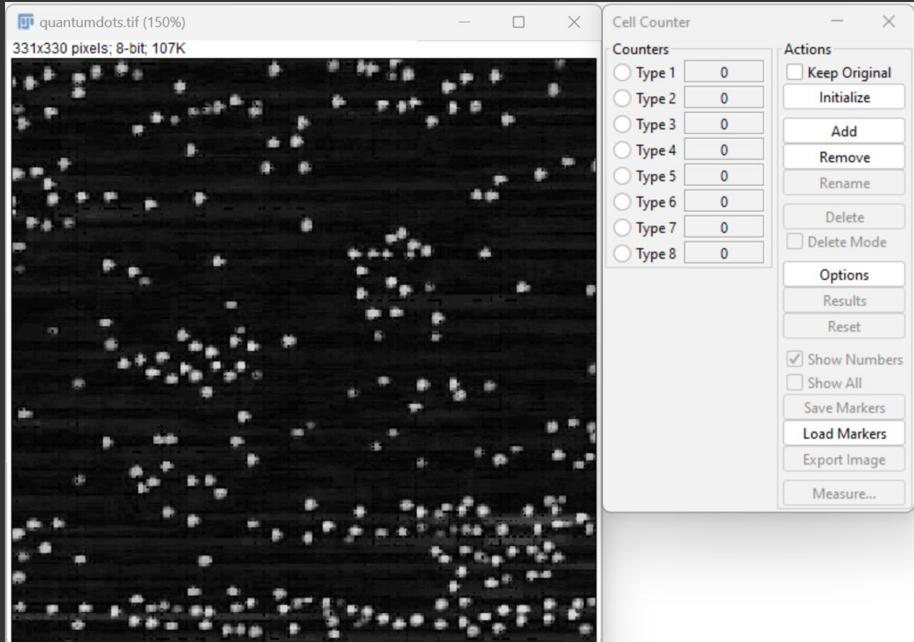
Plugins → Analysis → Cell Counter → Cell Counter

Initialize

Select a Type (e.g. Type 2)

User the cursor and then click - you get counts

You can use Delete to remove the last count



3 Measurements

3.5 ROI manager

File → Open samples → Hela Cells

Image → Duplicate (Channel 3, call it nuc)

Image → Lookup Tables → Gray

Image → Duplicate (Call it nuc-seg)

Image → Adjust → Threshold [Otsu, apply]

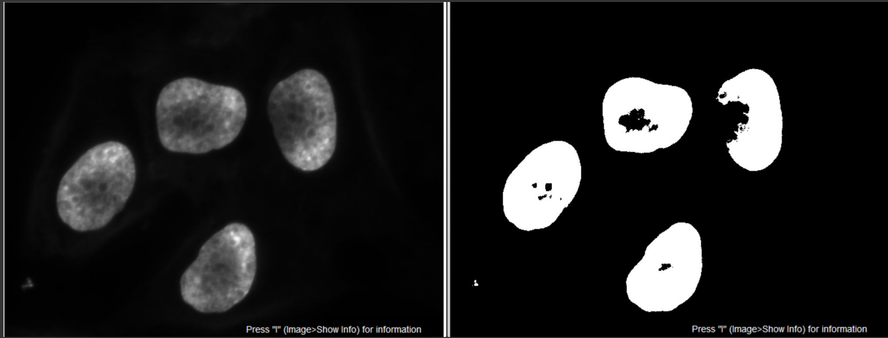
Analyze → Tools → ROI Manager

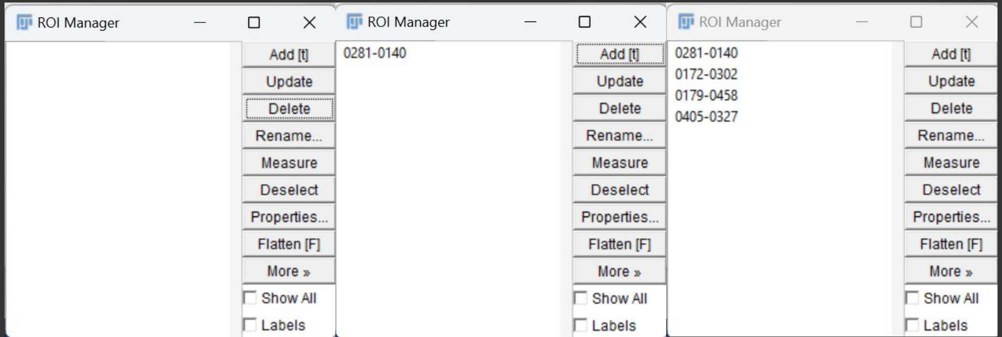
Wand → Select segmented object

ROI Manager → Add

Repeat

Try the Measure button on the manager





5 Multi-dimensional datasets

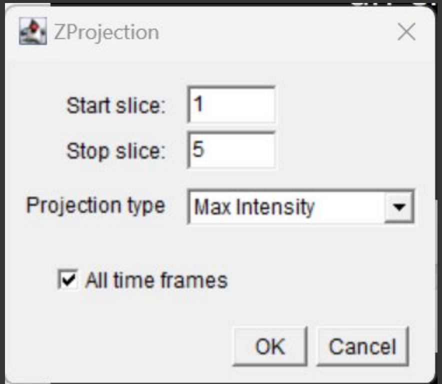
Fiji can handle up to 5-d data, color, x, y, z, time

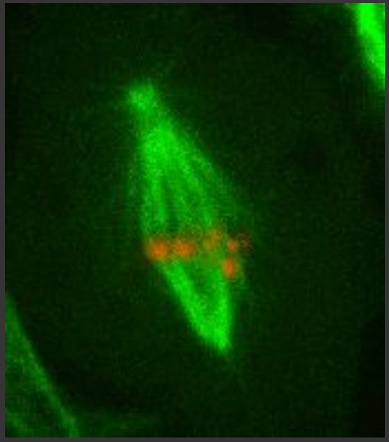
5.1 Z projections

File → Open samples → Mitosis

5d data is difficult to display in presentations, lets try a simple projection

Image → Stacks → ZProjection





5 Multi-dimensional datasets

5.2 Orthogonal views

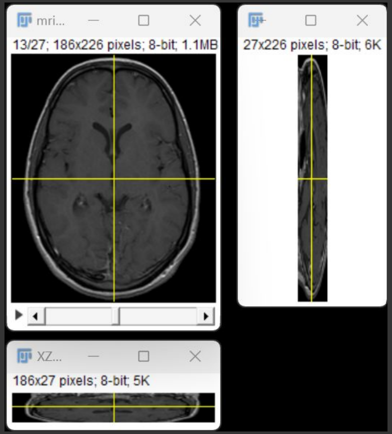
File → Open samples → MRI Stack

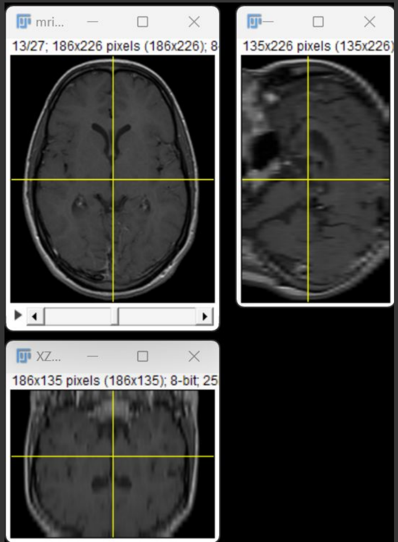
Image → Stacks → Orthogonal views

Image → Properties → Change voxel z to 5 for example

Image → Stacks → Orthogonal views

Try with EMBL → Samples → BBBC024c1n0_crop





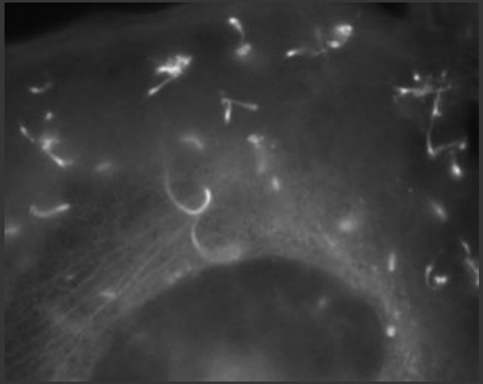
5 Multi-dimensional datasets

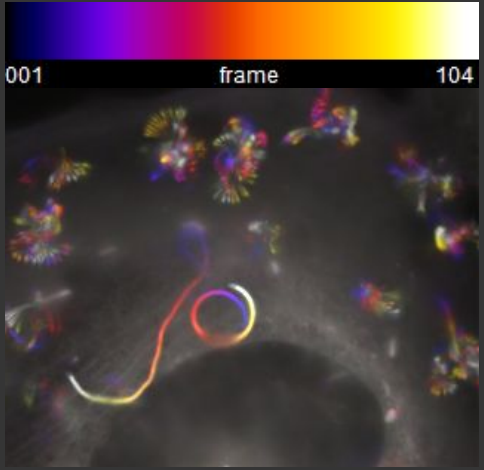
5.3 Time projections

EMBL → Samples → Listeria cells

Even time series can be difficult to display in presentations, lets try another projection

Image → Hyperstacks→ Temporal-Color Code





Classical image analysis workflow

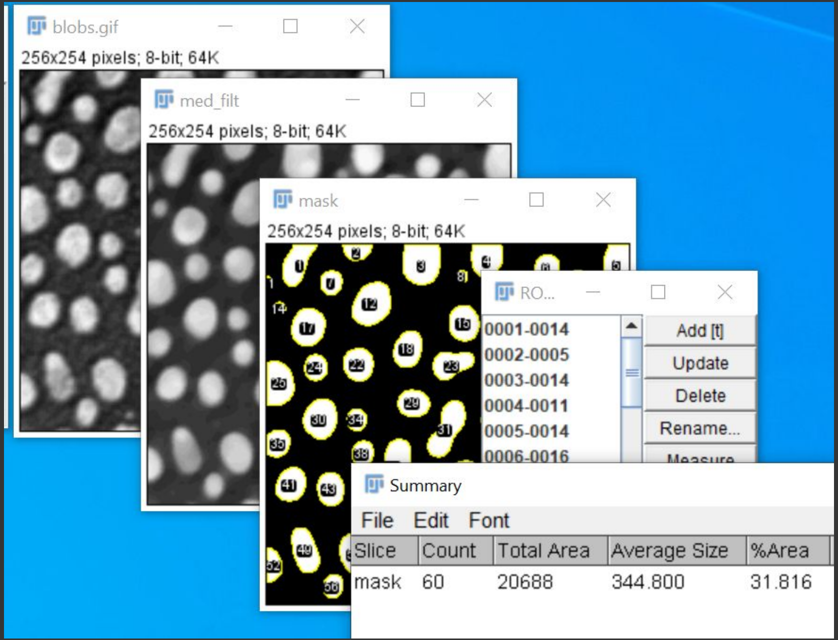
Image load

Pre-processing

Segmentation

Post processing

Quantification



6 Pre-processing

Background removal via filtering

File → Open → rice.tif : What are the challenges with this image

How can we remove non uniform illumination?

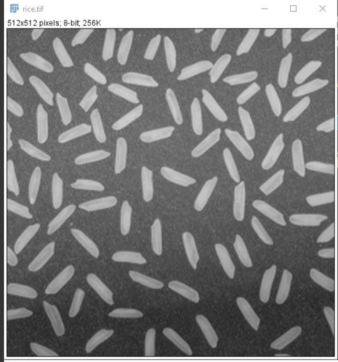
Image → Duplicate → Title: BG

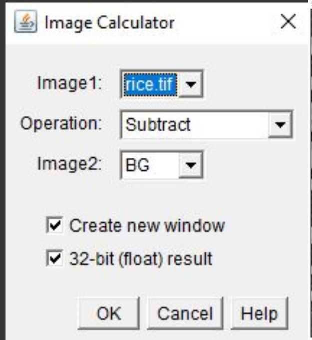
Process → Filters → Gaussian Blur

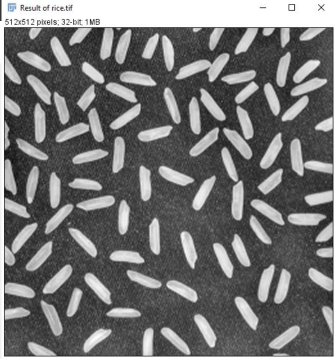
6.1 Find a sigma that gives you the background

6.2 Remove background from image

Process → Image Calculator







Interactive example of image filter kernels: https://setosa.io/ev/image-kernels/

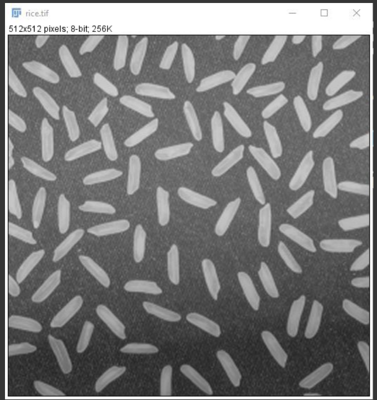
6 Pre-processing

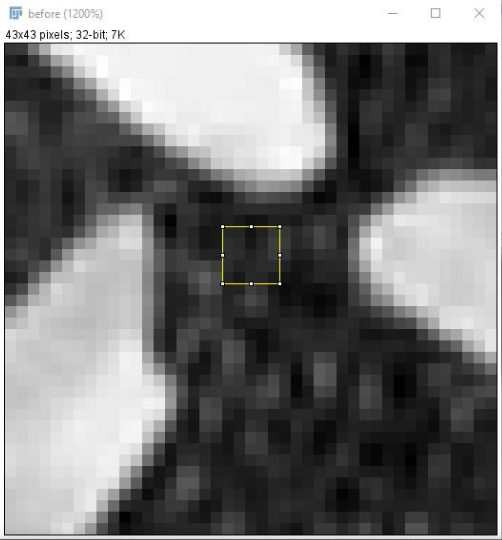
6.3 Further improve quality

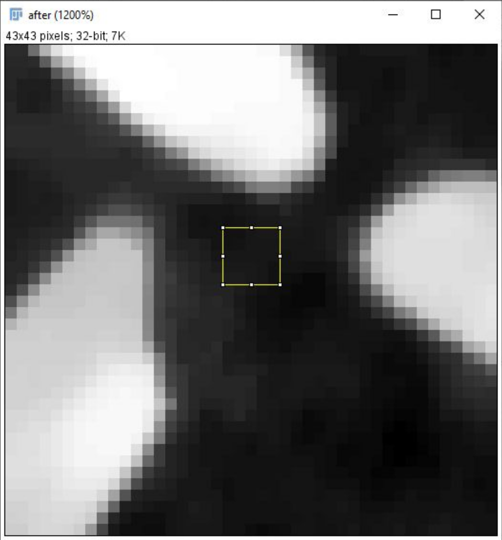
Rename results: Image → Rename → Title: bg free

Image → Duplicate → Title: Median filtered

Process → Filters → Median | Find a good radius







7 Segmentation

7.1 Try thresholding the original image

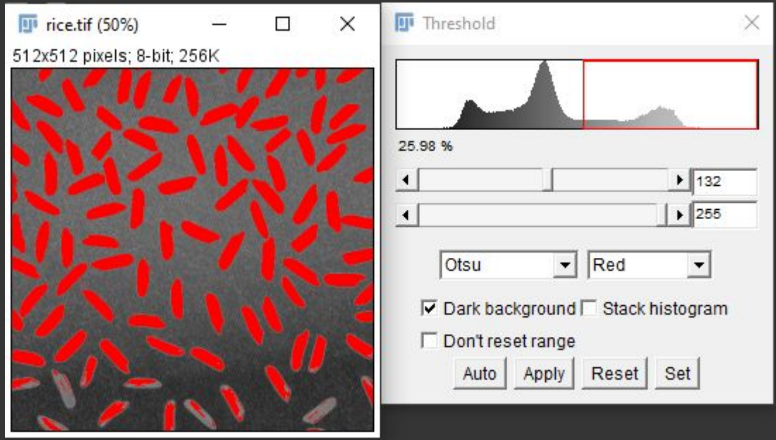
Select rice.tif

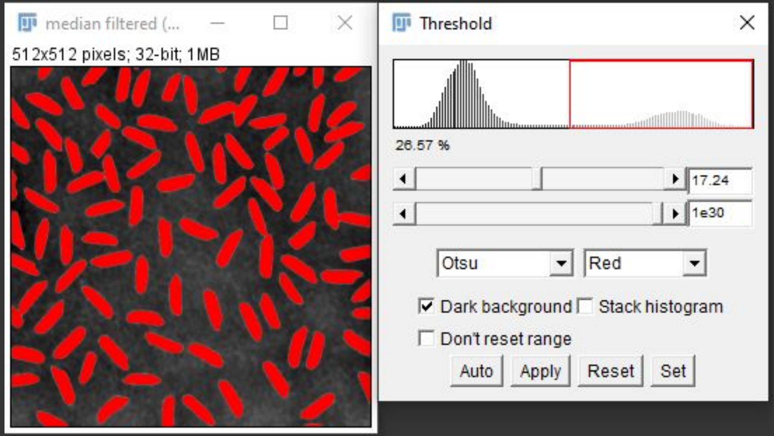
Image → Adjust → Threshold | Play with the menu

7.2 Try thresholding the processed image

Image → Adjust → Threshold | Play with the menu

Apply and convert to mask





8 Quantification

8.1 Let us count the rice

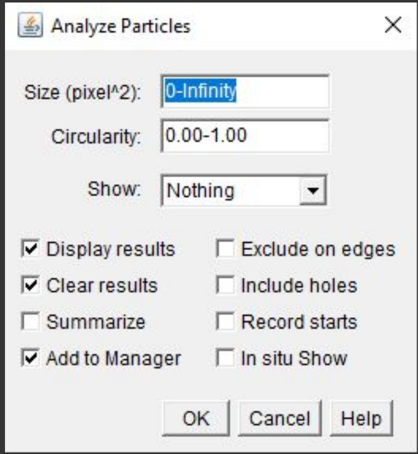
File → Open → rice_mask.tif

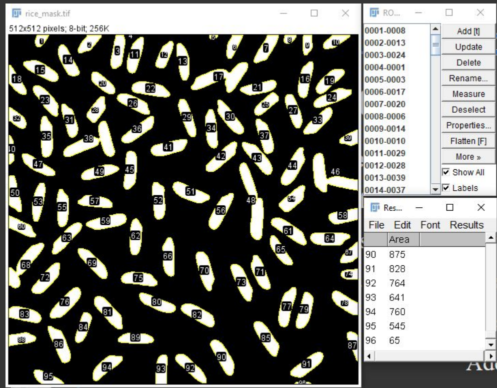
Analyze → Set measurements → Select: Area

Analyze → Analyze particles → Select: Display results,

Clear results

Add to Manager





How many rice grains do you get?

8.2 Exclude on edges - how many now?

8 Quantification

8.3 Simple histogram

Go to results table: Results → Distribution → Bins: 20

8.3 Do you see a problem?

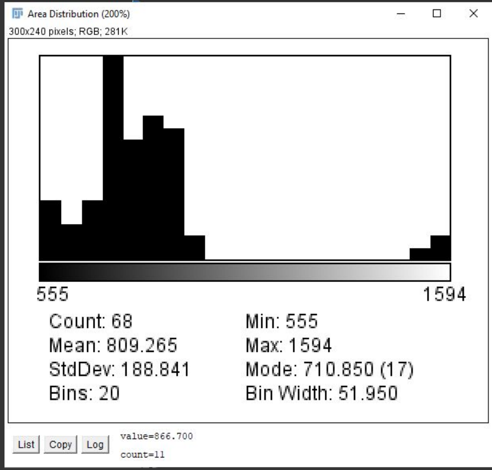
Try: Process → Binary → Watershed

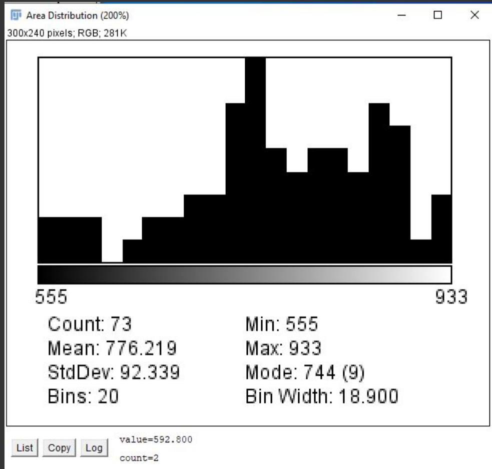
Analyze particles again

We don't have time to cover the theory of watershed.

Look at: https://imagej.net/Classic_Watershed

https://en.wikipedia.org/wiki/Watershed_(image_processing)





6 Pre-processing

6.5 Background removal - filtering in temporal dimension

EMBL → samples → listeriacells.tif (this will take a minute)

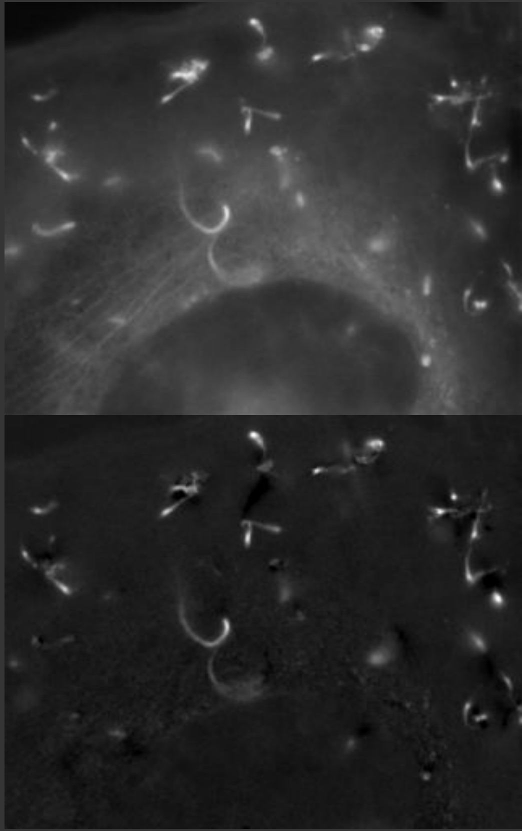
How can we remove non moving background

Image → Stack → Z project… → Median

Process → Image Calculator

You now should know what to do

Discuss: Why it works?



9 Segmenting with gradients

EMBL → samples → dic_cells.tif

9.1 try thresholding the image

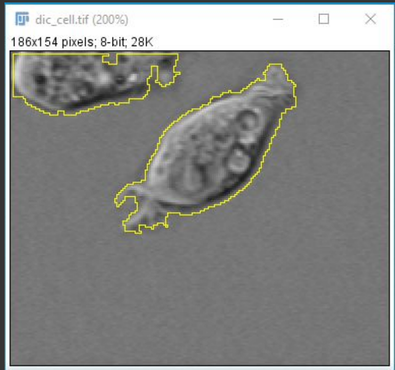
Image → Adjust → Threshold   Discuss: What are the problems?

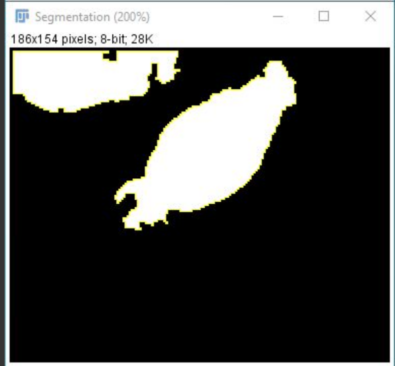
9.2 image gradient

Plugins → Process → Gradient (3D) → ok

9.3 thresholding the gradient image

Image → Adjust → Threshold → Method: Li





9 Segmenting with gradients

9.4 post processing - binary morphological operations

Process → Binary → Close

Process → Binary → Fill holes

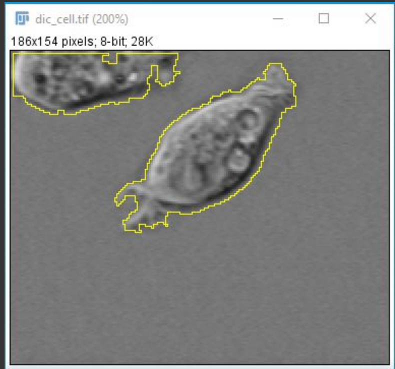
misc - Copy selection to original image

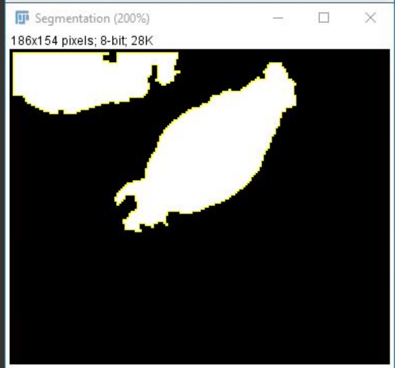
Edit → Selection → Create Selection

Edit → Selection → Restore Selection

Morphological operations are super important,

We don't have time, look at: https://imagej.net/MorphoLibJ





10 misc - Morphological operations

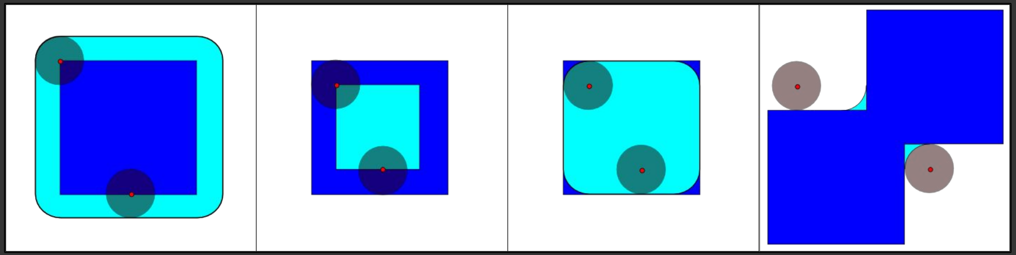
Morphological operations are super important,

We don't have time, sorry

Look at: https://imagej.net/MorphoLibJ

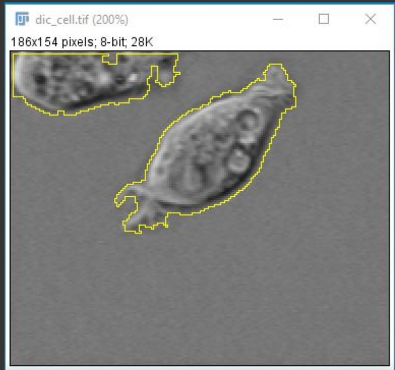
An the wiki-page: https://en.wikipedia.org/wiki/Mathematical_morphology

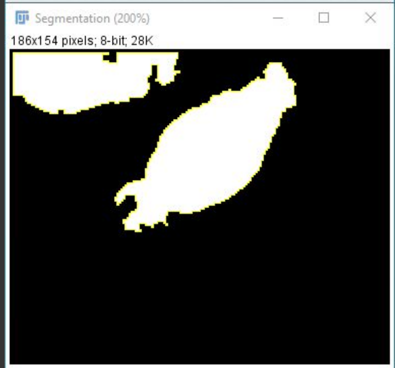
Dilation Erosion Open Close



Renatokeshet at the English Wikipedia

Author: Renato Keshet Source: Inkscape





11 Pre-processing

11.1 Filtering via Difference of Gaussians - DoG

EMBL ➝ samples ➝ eb1_8b.tif

How can we highlight spot/round like objects

DoG: Gauss blur im 1 - Gauss blur im 2; using: sigma_2 = 1.414 sigma_1

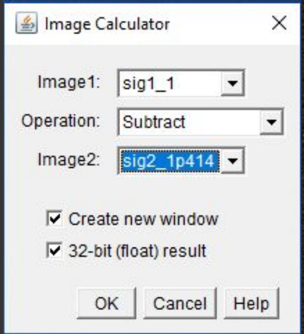
Image ➝ Duplicate ➝ Title: sig1_1

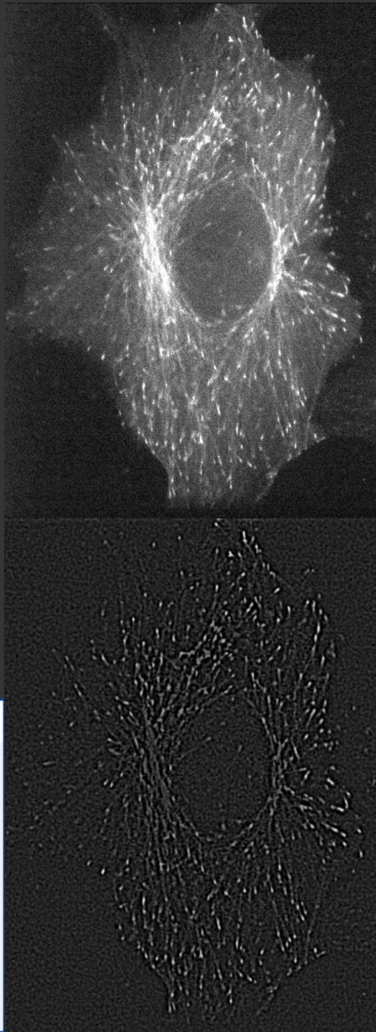
Process ➝ Filters ➝ Gaussian Blur ➝ Sigma: 1.0 ➝ Yes

Go to original: Image ➝ Duplicate ➝ Title: sig2_1p414

Process ➝ Filters ➝ Gaussian Blur ➝ Sigma: 1.414 ➝ Yes

Process ➝ Image Calculator ➝ sig_1 - sig_2





11 Pre-processing

11.2 Filtering via Difference of Gaussians - DoG

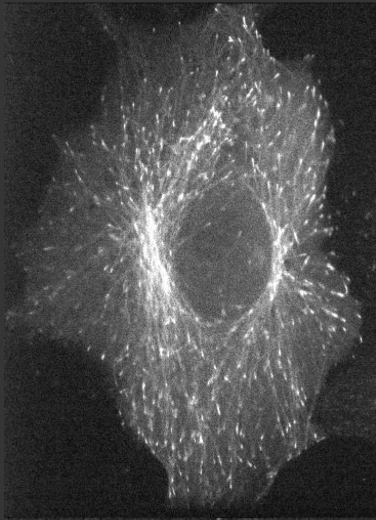
Again, sorry no time for theory today.

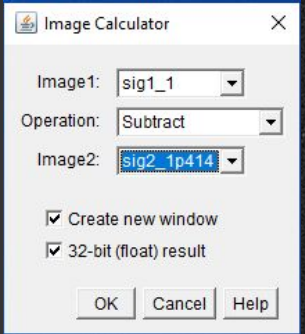
For DoG look at: https://en.wikipedia.org/wiki/Difference_of_Gaussians

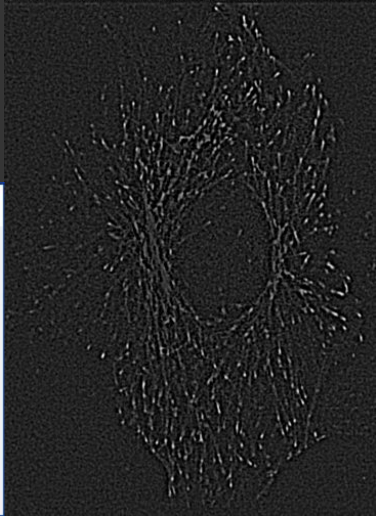
“Subtracting one image from the other preserves spatial information that lies between the range of frequencies that are preserved in the two blurred images. Thus, the DoG is a spatial band-pass filter that attenuates frequencies in the original grayscale image that are far from the band center.”

Think of it as a filter that enhances spots of a given size.

Larger the spot larger sigma has to be used!







12 misc - Plugins and update sites

5.2 Installation of plugins listed in update sites

Go to: https://imagej.github.io/list-of-update-sites/

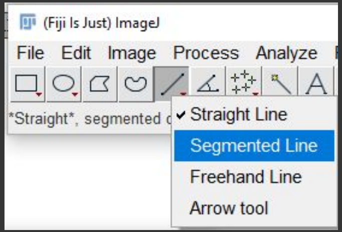
Lets us install the plug-in MorphoLibJ

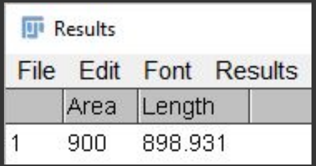
Help → Update → Manade update sites → IJPB-plugins → Close → Apply Changes

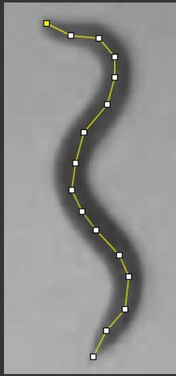
After restart you should now see the Plugins → MorphoLibJ tab

13 Length measurements

EMBL → samples → Celegans.png







13.1 Manual length measurement

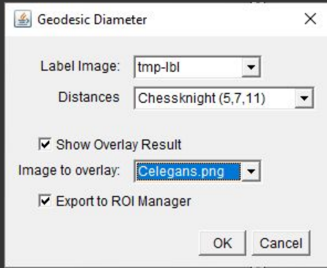
Annotate by hand using the segmented line

Analyse → Measure

13.2 Segment via Otsu

Image → Adjust → Threshold → Otsu, unselect “Dark background” → Apply

13.3 Geodesic Diameter



Plugins → MorphoLibJ → Dilate → Binary… → Connected Components Labeling

Plugins → MorphoLibJ → Analyze → Geodesic Diameter

Discus: Problems?

13 Length measurements

EMBL → samples → Celegans.png

13,4 Segment via Otsu

Image → Duplicate

Image → Adjust → Threshold → Otsu, unselect “Dark background” → Apply

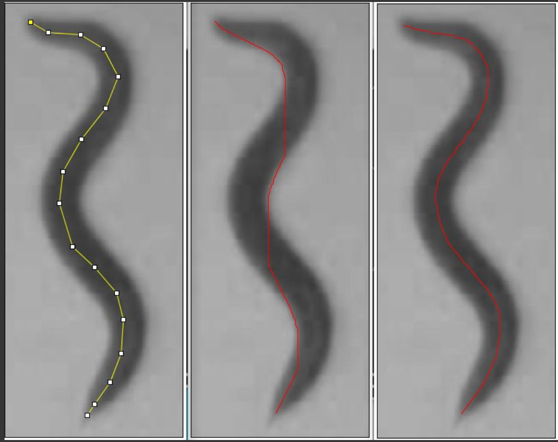
13.5 Geodesic Diameter with Skeletonized image

Process → Binary → Skeletonize

Process → Binary → Dilate

Plugins → MorphoLibJ → Dilate → Binary… → Connected Components Labeling

Plugins → MorphoLibJ → Analyze → Geodesic Diameter



Results

File Edit Font Results

	Area	Length
1	900	898.931

File Edit Font

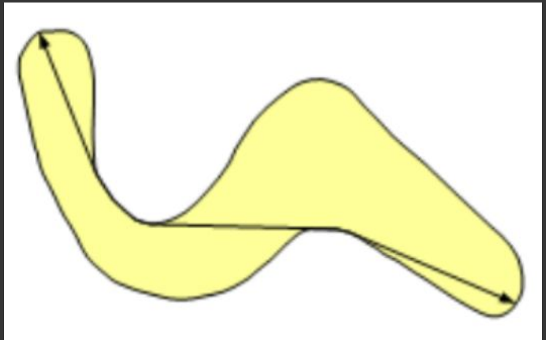
Label	GeodesicDiameter
1	888.614

13 Length measurements

Geodesic Diameter

From MorphoLibJ site we have:

"For particles with complex shapes, the geodesic diameter may be of interest. It corresponds of the largest geodesic distance between two points within a region, the geodesic distance being the length of the shortest path joining the two points while staying inside the region (Lantuejoul et al., 1981).



Lantuejoul, C. and Beucher, S. (Jan. 1981). "On the use of geodesic metric in image analysis". 121: 39-40. doi:10.1111/j.1365-2818.1981.tb01197.x.

Where to go from here?

- ➔ **Pete's book!**
  - https://bioimagebook.github.io/README.html
- ➔ **Neubias training resources**
  - https://neubias.github.io/training-resources/index.html
- ➔ **Find a problem and start working on it**
- ➔ **Check out the macro recorder**
  - https://github.com/CamachoDejay/Teaching-ImageJ-FIJI/tree/main/NorMIC/Day_2_Macro_recording

@camachodejay